

The Xbox... a Taste of Things to Come

The first time the GeForce3 made its appearance was not on a graphics card. Instead, it made its debut in Microsoft's gaming console—the Xbox. Boasting of steroidal specifications, this gaming console promised gameplay like nothing before it. At that time, the GeForce3

was code-named NV20 and it was one of the most powerful graphics sub-systems available for 3D computer graphics.



Gear up for the next generation of gaming with Microsoft's Xbox

Specifications of the Xbox	
CPU	733 MHz
Graphics processor (NV20)	250 MHz core
Total memory	64 MB
Memory bandwidth	6.4 GBps
Polygon performance	125 million polygons/sec
Pixel fill rate	4.0 Gigapixels/sec
Storage sub-system	5X DVD, 8 GB HDD

Until now, the easiest way to counter this problem was to add more memory and pump up the memory's frequency. However, with the GeForce2 Ultra's memory clocking at 460 MHz with bandwidths of up to 7.3 GBps, something else had to be done to alleviate the memory bottleneck. So the GeForce3 uses a couple of innovative approaches to access and use its onboard memory.

Splines: In the past, graphics applications used to send raw triangle data to the graphics card, which in turn would process this data and churn it out into displayable graphics. Nowadays, the complexity of the 3D elements in games reaches high levels and there are times when individual 3D elements are often composed of hundreds of thousands of polygons!

So instead of relying on raw triangle data to define complex 3D surfaces, they are now defined by a number of control points, also known as splines. This results in a significantly lower load on the memory bandwidth and greater efficiency is obtained, especially when working with complex 3D elements.

Crossbar memory controller: Another enhancement to the memory architecture is the crossbar memory controller. The GeForce3 uses four memory controllers that interact with each other and the GPU. Therefore, smaller chunks of information can be accessed and processed from many parts of the memory simultaneously.

HRAA (High Resolution Anti-Aliasing)

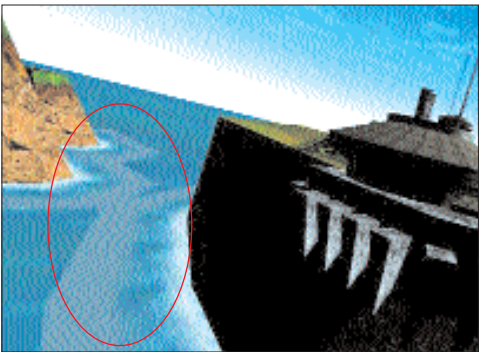
This is another feature that was incorporated to a lesser extent into the GeForce2 as well. Aliasing is the appearance of undesirable jagged edges in a 3D model when there are not enough pixels to represent it. The GeForce3 uses a process called Quincunx Sampling to minimise these 'jaggies' by interpolating pixels between the jagged edges. It does so by internally increasing the resolution of the game and then scaling it down to the required resolution. This feature is hardwired into the GeForce3; enabling HRAA to make edges and lines in 3D games appear smooth.

Lossless Z-compression

One of the most important components of the 3D processor is the Z-buffer. This is the area where the 3D processor stores depth information about a scene. The graphics processor uses this information to know which 3D objects lie in the foreground and which are in the background. Since this information is extensive in typical 3D games, it takes up a lot of memory bandwidth. To optimise the memory bandwidth, GeForce3 implements a form of lossless compression of this Z-buffer data and therefore allows the memory to be used more efficiently.

Z-Occlusion Culling

When a 3D object is hidden by another object in the foreground, there is in fact no need to draw that hidden part because it won't be visible to the viewer. If this approach is used, the information needed for drawing this object does not have to be passed to the graphics processor. The GeForce3 uses this technique of only processing graphical information that will eventually be drawn in the scene. This results in a significant increase in the efficiency of the graphics processor's throughput.



A scene without antialiasing (above) and with it (below). Notice the reduction in jaggies